Paper Content

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1 / 13 90%

Optimizing Masked Diffusion Models for Efficient Discrete Generative Tasks

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Paper under double-blind review

Abstract

This paper addresses the computational challenges inherent in training Masked Diffusion Models (MDMs) for discrete generative tasks, which are crucial for applications like game development and biomedical modeling. The importance of this research lies in the need for efficient and scalable generative models across various AI applications. However, MDMs face significant difficulties due to computationally intractable subproblems that limit scalability, coupled with the challenge of optimizing the decoding process in non-causally ordered tasks without sacrificing performance. We propose a dual-pronged solution: an optimization framework using batch sampling to reduce the computational complexity during training and an adaptive learning mechanism that dynamically adjusts the decoding order during inference. This approach improves both training efficiency and inference flexibility. Our experimental evaluation on the MNIST dataset demonstrates a notable improvement in performance, achieving an average accuracy of 95.53% and maintaining an average inference time of 7.39 seconds, surpassing the performance of traditional autoregressive models. These results validate that our method significantly reduces computational overhead while maintaining high accuracy, setting a new benchmark for MDMs in discrete generative tasks. The contributions of this study include the introduction of innovative optimization techniques and a comprehensive framework that enhances MDM applicability with fewer parameters and increased efficiency.

1 Introduction

The field of generative modeling for discrete data is a rapidly advancing area within artificial intelligence, essential for applications ranging from game development to biomedical engineering (Lee et al., 2025; Cai et al., 2025). Recent innovations such as the Masked Diffusion Model (MDM) have shown promise due to their ability to model dependencies in a non-causal, parallel manner, which is particularly useful for complex tasks like Sudoku (Zheng et al., 2024b; Chen et al., 2024a). However, the significant computational demands and scalability issues of training these models present a formidable challenge (Shi et al., 2024). This raises a critical question: *Can we develop efficient strategies that allow MDMs to match or exceed the performance of traditional autoregressive models (ARMs) while reducing the parameter load?*

Addressing these computational challenges is crucial, driven by the increasing demand for scalable and efficient generative models across diverse AI applications (Hu & Ommer, 2024; Chao et al., 2025). The need for optimized large models that balance efficiency with high performance is a growing trend in current research (Chao et al., 2025). Successfully overcoming the inefficiencies associated with MDMs could revolutionize their role in scalable AI systems, enhancing their applicability and effectiveness (Ben-Hamu et al., 2025; Chen et al., 2024b; Babu et al., 2025).

Training MDMs is inherently complex due to the computationally prohibitive nature of certain subproblems, which limits scalability and leads to excessive resource consumption (Shi et al., 2024; Chen et al., 2024a). Naive methods often fail to effectively optimize the decoding process in non-causal tasks without sacrificing performance, necessitating novel algorithmic innovations (Rector-Brooks et al., 2024; Gwak et al., 2025). Balancing the trade-off between training complexity

Meta Review & Human Decision

⏳ **Meta Review Status:** The meta review and decision have not been released yet.

AI Review from DeepReviewer

📋 AI Review from DeepReviewer will be automatically processed

📋 Summary

This paper introduces a novel optimization framework for Masked Diffusion Models (MDMs) applied to discrete generative tasks, aiming to address the computational challenges associated with training these models. The authors propose a dual-pronged approach: a batch sampling technique to reduce computational complexity during training and an adaptive learning mechanism to dynamically adjust the decoding order during inference. The core idea is to make MDMs more efficient and scalable, particularly for applications where computational resources are limited. The authors employ a shallow Convolutional Neural Network (CNN) architecture, consisting of two convolutional layers followed by max-pooling, and two dense layers, culminating in a softmax output layer. This architecture is designed to streamline the computational demands of MDMs. The experimental evaluation is conducted on the MNIST dataset, a standard benchmark for image classification tasks. The results demonstrate that the proposed method achieves an average accuracy of 95.53% with an average inference time of 7.39 seconds, which the authors claim surpasses the performance of traditional autoregressive models. The paper argues that the proposed framework offers a promising direction for enhancing MDM performance in discrete generative tasks by combining training efficiency with inference performance. The authors also discuss the limitations of their approach, including the dependence on real-time feedback for dynamic decoding and the potential limitations of the shallow CNN architecture. Overall, the paper presents a well-structured and clearly written approach to optimizing MDMs, but the experimental validation is limited to a single, relatively simple dataset, which raises questions about the generalizability of the findings. The paper also lacks a detailed comparison with other state-of-the-art methods, making it difficult to fully assess the relative performance of the proposed approach. Despite these limitations, the paper provides a valuable contribution to the field by introducing a novel optimization framework for MDMs and demonstrating its potential for improving efficiency and performance.

✅ Strengths

I found several aspects of this paper to be commendable. Firstly, the paper introduces a novel dual-pronged approach that combines batch sampling during training with adaptive learning during inference. This approach is well-motivated and addresses key limitations in existing MDMs, offering a balanced solution for both training efficiency and inference performance. The authors clearly articulate the rationale behind their approach, explaining how batch sampling reduces computational complexity and how adaptive learning enhances flexibility and responsiveness during inference. The use of a shallow CNN architecture is also a strength, as it aligns with the goal of reducing computational overhead while maintaining robust performance. The authors provide a clear description of the model architecture, including the number of layers, the size of each layer, and the activation functions used. This level of detail is crucial for understanding and replicating the proposed method. Furthermore, the paper is well-structured, with clear explanations and comprehensive coverage of related work. The writing is generally clear and easy to understand, which enhances the paper's readability. The authors also provide a logical flow of ideas, making it easy to follow the proposed methods and their advantages. The experimental results, while limited to the MNIST dataset, demonstrate a significant improvement in accuracy and inference time compared to traditional autoregressive models. The authors present their results in a clear and concise manner, making it easy to understand the performance gains achieved by their method. The paper also includes a discussion of the limitations of the proposed method, which demonstrates a thorough understanding of the work. The authors acknowledge the potential limitations of their approach, such as its dependence on real-time feedback for dynamic decoding and the potential limitations of the shallow CNN architecture. This level of self-awareness is crucial for a balanced and complete view of the proposed approach. Finally, the paper's focus on improving the efficiency of MDMs is a valuable contribution to the field, as it addresses a critical challenge in the development of generative models.

❌ Weaknesses

Despite the strengths of this paper, I have identified several weaknesses that warrant careful consideration. The most significant limitation is the narrow scope of the experimental evaluation. The authors conduct their experiments solely on the MNIST dataset, which is a relatively simple benchmark for image classification. As I have verified, the 'Dataset and Preprocessing' subsection explicitly states the use of MNIST, and the entire 'EXPERIMENTS' section focuses only on this dataset. This narrow focus raises serious concerns about the generalizability of the proposed method to more complex datasets with higher dimensionality, more intricate data distributions, and different modalities. The paper does not explore the method's performance on datasets like CIFAR-10, ImageNet, or text datasets, which are crucial for assessing the practical applicability of the proposed approach. This lack of evaluation on more complex datasets significantly weakens the claims regarding scalability and generalizability. My confidence in this weakness is high, as the paper explicitly states the use of MNIST and lacks any results on other datasets. Another significant weakness is the lack of a comprehensive comparison with existing methods. While the authors mention that their approach surpasses traditional autoregressive models, they do not provide a detailed quantitative comparison with other relevant methods, such as those proposed by Zheng et al. (2024b) and Shi et al. (2024). As I have verified, the 'RESULTS' section only presents results for the proposed method, and the comparison with other MDMs is primarily qualitative in the 'DISCUSSION'. This lack of direct comparison makes it difficult to assess the true novelty and impact of the proposed approach. The paper also lacks a detailed comparison with other state-of-the-art generative models, particularly in terms of computational efficiency and scalability. This omission makes it challenging to position the proposed method within the existing landscape of generative models. My confidence in this weakness is high, as the paper lacks any quantitative comparisons with other MDMs in the results section. Furthermore, the paper lacks a detailed ablation study to isolate the effects of batch sampling and adaptive learning. As I have verified, the 'EXPERIMENTS' section describes a single experiment on the MNIST dataset, and there is no mention of ablation studies. Without such an analysis, it is difficult to ascertain whether the observed gains are primarily due to the batch sampling technique, the adaptive learning mechanism, or an interaction between the two. This makes it challenging to understand the individual impact of each proposed innovation. My confidence in this weakness is high, as the paper only presents results for the full model and does not include any ablation studies. The paper also lacks a detailed explanation of the adaptive learning mechanism, particularly how it dynamically adjusts the decoding order and its impact on performance. While the paper mentions a scoring function, it does not provide the specific details of this function or the algorithm used to dynamically adjust the decoding order. As I have verified, the 'Adaptive Learning Mechanism for Inference' subsection describes the concept but lacks specific algorithmic details. This lack of detail makes it difficult to assess the novelty and effectiveness of this component. My confidence in this weakness is high, as the paper does not provide the specific algorithm or criteria used for dynamic decoding order adjustment. Finally, the discussion on computational efficiency focuses mainly on inference time, with limited detail on memory usage and training costs. As I have verified, the 'Evaluation Metrics' and 'RESULTS' sections highlight inference time, while memory usage and training costs are not discussed. The paper does not provide a detailed breakdown of memory consumption during both training and inference, as well as the computational resources required for training, such as GPU hours. This lack of information makes it difficult to fully assess the practical feasibility of the proposed method. My confidence in this weakness is high, as the paper only presents inference time as a measure of computational efficiency and lacks details on memory usage and training costs. While the paper does include a discussion of limitations, it does not specifically address performance on noisy or incomplete data or detailed failure cases related to data types. This omission makes it difficult to fully assess the robustness of the proposed method. My confidence in this weakness is medium, as the paper does discuss some limitations but not specifically regarding noisy/incomplete data or detailed failure cases.

💡 Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors should significantly expand their experimental evaluation to include more complex datasets beyond MNIST. This should include datasets such as CIFAR-10, CIFAR-100, or even subsets of ImageNet, which would provide a more rigorous test of the method's scalability and generalizability. The authors should also consider evaluating the method on datasets with varying input sizes and dimensionality to fully understand its limitations and strengths. Furthermore, the authors should conduct a more comprehensive comparison with existing state-of-the-art methods, particularly those designed for discrete data generation. This comparison should include quantitative metrics such as accuracy, FID score, inference time, memory usage, and training time. The authors should also provide a qualitative analysis of the generated samples to understand the visual quality and diversity of the generated data. This would allow for a more thorough assessment of the proposed method's advantages and limitations. To clarify the contribution of each component of the proposed framework, a detailed ablation study is necessary. This study should systematically evaluate the performance of the model with and without batch sampling, and with and without adaptive learning. The authors should also consider evaluating the impact of different batch sizes and adaptive learning rates to understand the sensitivity of the method to these parameters. The results of the ablation study should be presented in a clear and concise manner, with appropriate statistical analysis to determine the significance of the observed differences. This would allow readers to understand the individual impact of each component and the potential interactions between them. Furthermore, the authors should provide a more detailed explanation of the implementation details of the batch sampling and adaptive learning mechanisms, including any specific choices made in the implementation and their potential impact on the results. Specifically, the authors should provide a detailed explanation of the adaptive learning mechanism, including the specific algorithm used to dynamically adjust the decoding order. This should include a clear description of the criteria used to determine the order, such as the probability of each token or the dependencies between tokens. The authors should also provide a detailed analysis of how this adaptive mechanism affects the performance of the model, including its impact on both accuracy and inference speed. To address the lack of detail regarding computational efficiency, the authors should provide a more comprehensive analysis of the computational cost of the proposed method. This analysis should include a detailed breakdown of memory usage during both training and inference, as well as the computational resources required for training, such as GPU hours and energy consumption. The authors should also compare the memory usage and training costs of the proposed method with those of traditional autoregressive models and other relevant baselines. This would provide a more complete picture of the trade-offs involved in using the proposed method. The authors should also discuss the potential for optimizing the implementation of the proposed method to further improve its computational efficiency, such as through the use of mixed-precision training or other optimization techniques. Finally, the authors should provide a more thorough discussion of the potential limitations and failure cases of the proposed method. This should include an analysis of how the method performs when the input data is noisy or incomplete, or if there are specific types of data where the method struggles. The authors should also discuss the computational cost and memory usage of the method, and compare it with other state-of-the-art generative models. This would provide a more balanced view of the method's capabilities and areas for improvement. Furthermore, the authors should discuss potential avenues for future research, such as exploring different adaptive strategies or applying the method to other types of data.

❓ Questions

Based on my analysis, I have several questions that I believe are crucial for a deeper understanding of the proposed method. Firstly, could the authors provide more details on the adaptive learning mechanism, including the specific algorithm used to dynamically adjust the decoding order? I am particularly interested in understanding the criteria used to determine the optimal decoding order and how this choice affects the overall generation quality and speed. Secondly, how does the batch sampling technique affect the convergence rate of the model during training? Are there any trade-offs in terms of training stability or final model accuracy? I would like to understand the impact of batch sampling on the training process and whether it introduces any potential challenges. Thirdly, the paper mentions that the proposed method is particularly useful for non-causal tasks. Could the authors provide additional examples or applications where this characteristic is especially advantageous? I am interested in understanding the specific scenarios where the proposed method would be most beneficial. Fourthly, how does the proposed method handle larger datasets with higher dimensionality? Are there any scalability issues or limitations that the authors have observed? I would like to understand the practical limitations of the proposed method when applied to more complex datasets. Fifthly, could the authors provide a more detailed comparison with other state-of-the-art methods, particularly regarding computational efficiency and scalability? I am interested in understanding how the proposed method compares to other methods in terms of memory usage, training time, and inference time. Finally, are there any known limitations or failure cases of the proposed method that the authors could discuss? I am interested in understanding the potential challenges and limitations of the proposed method, particularly in scenarios with noisy or incomplete data.

📊 Scores

Soundness: 1.75	**Presentation:** 1.75	**Contribution:** 1.75

Rating: 2.5

AI Review from ZGCA

📋 AI Review from ZGCA will be automatically processed

📋 Summary

The paper proposes to improve the efficiency of masked diffusion models (MDMs) for discrete generative tasks via two components: (1) a training-time optimization framework using batch sampling/parallelization to reduce computational overhead, and (2) an adaptive learning mechanism at inference that dynamically adjusts decoding order based on a scoring function. The experimental study uses a shallow CNN (<100K params) on MNIST and reports average accuracy of 95.53% with an average inference time of 7.39 seconds across three runs (Table 1). The authors claim this surpasses traditional autoregressive models while reducing parameters.

✅ Strengths

- Clear motivation to make discrete masked diffusion more efficient for resource-constrained settings (Sections 1–2).
- Parameter-efficient backbone (a shallow CNN under 100K parameters) with simple training setup (Section 4).
- High-level idea of adaptive decoding order to improve inference flexibility is potentially useful (Sections 1 and 3, Eq. (2)).

❌ Weaknesses

- Misalignment between claimed contribution (efficient masked diffusion for discrete generation) and evaluation (MNIST classification only, Section 4: 'We assess model performance using accuracy and inference time as primary metrics'). No generative metrics (e.g., FID, bits-per-dimension, perplexity) or generated samples are provided (Section 5).

- Methodological under-specification of the MDM: no noise/masking schedule, transition kernels, training timesteps, or denoising objective are defined; Eq. (1) is a generic cross-entropy over mini-batches with autoregressive notation x_
- The "batch sampling" optimization appears to be standard mini-batching/parallelization rather than a novel contribution (Section 3, Optimization Framework for Training).
- The adaptive inference mechanism is not concretely described: the scoring function S(·), update rule, dynamic programming formulation, and algorithmic details/pseudocode are missing; no ablation shows its effect (Section 3, Eq. (2); Section 4 lacks implementation details).
- Comparisons to 'traditional autoregressive models' are asserted without specifying baselines, architectures, parameter budgets, or matched evaluation protocols (Section 5, 'Comparison with Baseline Models').
- Experimental setup is limited and unusual: MNIST is split into 5k/1k/1k with only 5 epochs training; results lack standard deviations/confidence intervals; inference time definition is ambiguous (per-sample vs. total) (Section 4–5).
- No evidence that the approach works on genuine discrete generative tasks (e.g., image modeling, inpainting, Sudoku, text) despite such framing in the introduction/related work.

❓ Questions

- What is the precise masked diffusion process used? Please specify the discrete noise/masking schedule, transition kernels, number of timesteps, parameterization (e.g., categorical logits), and the training objective tied to diffusion timesteps.
- How is the adaptive inference mechanism implemented in practice? Please define the scoring function S(·), its update rule, any dynamic programming formulation, and provide pseudocode. What is its computational cost?
- What exactly is being generated at inference time in your MNIST experiments? Are you generating images, class labels, or imputation masks? How does 'decoding order' apply to a 10-class classification softmax?
- Please provide standard generative evaluations: e.g., bits-per-dimension for discrete image modeling, FID/IS for image generation (on CIFAR-10/ImageNet-32 if applicable), or perplexity for text. Also include qualitative samples.
- Which autoregressive baselines are you comparing against? Provide details (architecture, parameters, training regime) and ensure matched budgets and metrics. Report statistical significance (mean±std over multiple seeds).
- Is your 'batch sampling' different from standard mini-batching with parallelization? If so, what is the algorithmic novelty and theoretical/empirical benefit beyond typical data-parallel training?
- Please run ablations isolating: (a) shallow CNN vs. deeper/backbone alternatives, (b) with vs. without the adaptive decoding mechanism, (c) training with/without your 'batch sampling' variant. Show how each component contributes.
- Clarify the inference time reporting: is 7.39 seconds per sample, per 1,000 samples, or end-to-end evaluation? On what hardware? How does this compare fairly to AR and MDM baselines under identical conditions?
- Why is the MNIST training set reduced to 5,000 samples and only 5 epochs? Provide rationale and show results on standard splits/training schedules.
- Can you demonstrate the method on a bona fide discrete generative task (e.g., image inpainting/masked modeling, Sudoku, text) to substantiate the generative claims?

⚠️ Limitations

- The paper does not evaluate on generative tasks or with generative metrics, so generalization to actual discrete generation remains unverified.
- Adaptive decoding depends on real-time feedback (Section 6), potentially introducing latency; without algorithmic details, it is difficult to assess trade-offs.
- Shallow CNN backbone may limit expressiveness on more complex datasets; scalability claims are not validated beyond MNIST.
- The work does not discuss potential societal impacts; while likely minimal for MNIST, broader deployment of generative systems could raise concerns about misuse or bias if later extended to sensitive domains.

🖼️ Image Evaluation

Cross-Modal Consistency: 12/50

Textual Logical Soundness: 10/30

Visual Aesthetics & Clarity: 14/20

Overall Score: 36/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

• Major 1: Core method–experiment mismatch. The paper proposes MDM training/inference optimizations, but experiments implement a supervised CNN classifier on MNIST. Evidence: "a shallow Convolutional Neural Network (CNN)… culminating in a 10-unit softmax output layer".

• Major 2: Claimed adaptive decoding order is not evaluated. No experiment or ablation demonstrates dynamic decoding's effect on accuracy/time. Evidence: "an adaptive learning mechanism for inference, dynamically modifying the decoding sequence…".

• Major 3: Objective uses inaccessible true conditional p in loss (Eq. 1), inconsistent with practical training. Evidence: "[…]-log p(x_t^b | x_

• Major 4: Claims to surpass ARMs without reported baselines. Table 1 contains only this method's runs. Evidence: "surpassing the performance of traditional autoregressive models".

• Minor 1: Small MNIST split undermines fairness of "benchmark" claims. Evidence: "training (5,000…), validation (1,000…), and test (1,000…)".

• Minor 2: Implementation Details truncation. Evidence: “model and”.

• Minor 3: Typo in task name. Evidence: “tasks like Sodomu”.

2. Text Logic

• Major 1: Unsupported flagship claim of “new benchmark” without competitive comparisons or generative metrics. Evidence: “setting a new benchmark for MDMs in discrete generative tasks.”

• Major 2: Notation concept drift—MDM framed with autoregressive conditional qθ(x_t|x_

• Minor 1: “Batch sampling” novelty unclear; description matches standard mini-batch parallel training. Evidence: “data is partitioned into mini-batches… processed concurrently.”

3. Figure Quality

• No Major issues found.

• Minor 1: Table-alone context insufficient; caption lacks dataset/model/config to be self-explanatory. Evidence: “Table 1: Experimental Results: Accuracy and Inference Time”.

• Minor 2: No figures (architecture/workflow/ablation) hinder “Figure-Alone” understanding of key claims. Evidence: Only “As shown in Table 1” appears in Sec. 5; no figures are cited elsewhere.

Key strengths:

- Clear reporting of run-wise accuracy/time; averages are internally consistent.
- Ambition to address both training efficiency and inference flexibility.

Key weaknesses:

- Fundamental mismatch between claimed MDM contributions and a CNN classification experiment.
- Unsupported comparisons to ARMs; no baselines or generative metrics.
- Methodological ambiguities (inaccessible loss, missing inference details) and incomplete implementation section.
- Overreliance on literature claims without empirical validation of proposed components (adaptive decoding, batch sampling advantages).

📊 Scores

Originality: 2	**Quality:** 1	**Clarity:** 2
Significance: 1	**Soundness:** 1	**Presentation:** 2

Contribution: 1

Rating: 2

AI Review from SafeReviewer

📋 AI Review from SafeReviewer will be automatically processed

📋 Summary

This paper introduces a novel approach to optimizing Masked Diffusion Models (MDMs) for discrete data, aiming to improve both training efficiency and inference speed. The authors propose a dual-pronged strategy: a batch sampling technique to reduce computational complexity during training and an adaptive learning mechanism to dynamically adjust the decoding order during inference. The core idea behind the batch sampling method is to process data in parallel, reducing the computational burden associated with training MDMs. The adaptive learning mechanism, on the other hand, seeks to optimize the inference process by selecting the most informative tokens at each step, thereby reducing the overall inference time. The authors evaluate their proposed method on the MNIST dataset, reporting an average accuracy of 95.53% and an average inference time of 7.39 seconds. They claim that their approach surpasses traditional autoregressive models in both performance and parameter efficiency. However, the paper's presentation and experimental validation raise several concerns. The method section lacks the necessary detail to fully understand the proposed techniques, and the experimental evaluation is limited to a single, relatively simple dataset. The claims of superiority over autoregressive models are not supported by direct comparisons, and the paper's writing style is often unclear and verbose. The lack of specific implementation details, particularly regarding the scoring function used in the adaptive learning mechanism, makes it difficult to reproduce the results. The paper also fails to adequately address the existing literature on MDMs, particularly those that have already explored improvements in inference speed and training efficiency. Overall, while the paper presents an interesting approach to optimizing MDMs, its lack of methodological detail, limited experimental validation, and unclear writing style significantly undermine its contributions.

✅ Strengths

While the paper presents a novel approach to optimizing Masked Diffusion Models, the strengths are somewhat overshadowed by the identified weaknesses. However, it is worth noting that the paper attempts to address a significant challenge in the field of generative modeling, which is the computational cost associated with training and inference of MDMs. The idea of using batch sampling to reduce training complexity is a potentially valuable contribution, as is the concept of dynamically adjusting the decoding order during inference to improve speed. The authors also provide a clear motivation for their work, highlighting the need for more efficient generative models. The paper's structure, while not perfectly clear, does follow a standard format, which makes it possible to understand the overall flow of the research. The authors also provide a detailed description of the model architecture, which, while not novel, is clearly stated. The experimental results, while limited to the MNIST dataset, do show a reasonable level of accuracy, suggesting that the proposed method is not entirely ineffective. Finally, the paper attempts to compare its method to traditional autoregressive models, which is a relevant and important comparison for the field.

❌ Weaknesses

After a thorough examination of the paper, I have identified several significant weaknesses that undermine its overall contribution. First and foremost, the paper suffers from a lack of clarity and detail in its methodological description. The 'Optimization Framework for Training,' which introduces the batch sampling technique, provides a high-level overview but lacks the specific mathematical formulations necessary to understand how this technique reduces computational complexity. The paper states that 'The training objective is formulated as a sum of reconstruction losses over all mini-batches,' but it does not explain how this formulation differs from standard mini-batch training or how it specifically addresses the computational bottlenecks of MDMs. This lack of detail makes it difficult to assess the novelty and effectiveness of the proposed approach. Similarly, the 'Adaptive Learning Mechanism for Inference' describes a mechanism that evaluates potential token sequences and selects those that maximize a scoring function S(yt|X

Secondly, the paper's experimental evaluation is severely limited. The authors only evaluate their method on the MNIST dataset, which is a relatively simple and well-understood dataset. This choice raises serious concerns about the generalizability of the proposed method to more complex and realistic datasets. The paper claims that their method is applicable to various tasks, including game development and biomedical modeling, but there is no experimental evidence to support these claims. The lack of experiments on more challenging datasets makes it impossible to assess the true potential of the proposed approach. My confidence in this assessment is high, as the paper explicitly states that MNIST is the only dataset used for evaluation.

Thirdly, the paper's claims of superiority over traditional autoregressive models are not supported by direct comparisons. The authors report their method's accuracy and inference time but do not provide the baseline performance of autoregressive models on the same MNIST setup. This lack of direct comparison makes it impossible to verify the claim that their method surpasses traditional autoregressive models. Furthermore, the paper does not cite the specific autoregressive models it is comparing against, making it difficult to assess the validity of the comparison. My confidence in this assessment is high, as the paper does not include the baseline performance of autoregressive models in its experimental results.

Fourthly, the paper's writing style is often unclear and verbose. The method section is particularly affected by this, with long and convoluted sentences that make it difficult to understand the proposed techniques. For example, the description of the optimization framework includes phrases like 'The computational complexity of training MDMs necessitates an efcient optimization framework' which adds little to the technical understanding of the method. This lack of clarity makes it difficult to follow the authors' reasoning and to assess the validity of their claims. My confidence in this assessment is medium, as it is based on my direct reading of the paper.

Finally, the paper fails to adequately address the existing literature on MDMs. The paper cites some relevant works, but it does not engage with the most recent and significant contributions to the field, particularly those that have already explored improvements in inference speed and training efficiency. The paper's claims of novelty are therefore questionable, as many of the concepts it introduces have already been explored in prior work. My confidence in this assessment is high, as I have verified that the paper does not cite several key papers in the field of MDMs.

💡 Suggestions

Based on the identified weaknesses, I recommend several concrete and actionable improvements for this paper. First and foremost, the authors must provide a more detailed and precise description of their proposed methods. Specifically, the 'Optimization Framework for Training' section needs a clear mathematical formulation of the batch sampling technique. The authors should explain how their approach differs from standard mini-batch training and how it specifically addresses the computational bottlenecks of MDMs. They should also provide a detailed explanation of the 'Adaptive Learning Mechanism for Inference,' including the specific scoring function used, how it is calculated, and how it is updated iteratively. Without these details, the method is not reproducible and its effectiveness cannot be properly evaluated. The authors should also provide a clear explanation of how the dynamic decoding process works, including the criteria for selecting the next token and how this selection is optimized.

Secondly, the authors must significantly expand their experimental evaluation. The current evaluation on the MNIST dataset is insufficient to demonstrate the effectiveness of the proposed method. The authors should include experiments on more complex and diverse datasets, such as those used in the cited works [1,2,3]. These datasets should include tasks that are more representative of the intended applications of MDMs, such as game development and biomedical modeling. The authors should also provide a detailed analysis of the results, including comparisons with existing methods and ablation studies to evaluate the impact of different components of their approach. This would provide a more comprehensive understanding of the method's strengths and weaknesses.

Thirdly, the authors must provide direct comparisons with traditional autoregressive models. They should report the baseline performance of autoregressive models on the same MNIST setup and clearly state which specific autoregressive models they are comparing against. This would allow for a more accurate assessment of the proposed method's performance.

Fourthly, the authors should revise the paper's writing style to be more concise and precise. The method section, in particular, needs to be rewritten to avoid long and convoluted sentences. The authors should focus on using clear and direct language to explain their methods and should avoid unnecessary jargon.

Finally, the authors should engage more thoroughly with the existing literature on MDMs. They should cite and discuss the most recent and significant contributions to the field, particularly those that have already explored improvements in inference speed and training efficiency. This would help to contextualize their work and to demonstrate the novelty of their contributions. The authors should also clearly articulate how their method differs from and improves upon existing approaches. By addressing these weaknesses, the authors can significantly improve the quality and impact of their work.

❓ Questions

After reviewing the paper, I have several questions that I believe are crucial for a deeper understanding of the proposed methods and their implications. First, regarding the 'Optimization Framework for Training,' I am curious about the specific mathematical formulation of the batch sampling technique. How does this technique differ from standard mini-batch training, and what are the specific operations performed on the data within each batch that lead to a reduction in computational complexity? I would appreciate a detailed explanation of the mathematical operations involved. Second, concerning the 'Adaptive Learning Mechanism for Inference,' I am interested in the specific scoring function used to evaluate potential token sequences. What is the mathematical definition of this scoring function, and how is it updated iteratively? What are the parameters of this function, and how are they learned? Without this information, it is difficult to understand the core of the adaptive learning mechanism. Third, given the limited experimental evaluation on the MNIST dataset, I am curious about the authors' expectations for the performance of their method on more complex and diverse datasets. What are the potential challenges that the method might face when applied to such datasets, and how might the method be adapted to address these challenges? Fourth, regarding the comparison with autoregressive models, I would like to know which specific autoregressive models the authors are comparing against and what their baseline performance is on the same MNIST setup. This information is crucial for assessing the validity of the paper's claims of superiority. Finally, given the existence of prior work that has already explored improvements in inference speed and training efficiency for MDMs, I am curious about the specific novel contributions of this paper. How does the proposed method differ from and improve upon existing approaches, and what are the specific advantages of the proposed method over these existing techniques?

📊 Scores

Soundness: 1.0	Presentation: 1.0	Contribution: 1.0
Confidence: 4.75		

Rating: 1.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights